



DEVELOPMENT

Pump it Up: Data Mining the Water Table

Can you predict which water pumps are faulty to promote access to clean, potable water across Tanzania? This is an intermediate-level practice competition.

**Intermediate** practice**1 year** left**19,827** joined

Compete! →

Navigation

Home

About Taarifa

Problem description

Official rules

Leaderboard

Challenge Summary

Can you predict which water pumps are faulty?

Using data from [Taarifa](#) and the [Tanzanian Ministry of Water](#), can you predict which pumps are functional, which need some repairs, and which don't work at all? This is an intermediate-level practice competition. Predict one of these three classes based on a number of variables about what kind of pump is operating, when it was installed, and how it is managed. A smart understanding of which waterpoints will fail can improve maintenance operations and ensure that clean, potable water is available to communities across Tanzania.

This competition is for learning and exploring, so the deadline may be extended in the future.

How to compete

1. Click the "Join the competition" button on the sidebar to enroll in the competition.
2. Get familiar with the problem on the Problem Description and About page.
3. Download the data from the Data tab.
4. Create and train your own model.
5. Use your model to generate predictions that match the submission format. Click "Submit" in the sidebar, and then "Make new submission". You're in!
6. Bonus: share your work! Visit the "Share your work" page after joining and add a link to your approach.

Pump diagram courtesy of [Manco Capac via Wikimedia Commons](#) (CC BY-SA 3.0)

We use data science & AI to help mission-driven organizations offer more impactful services.

ABOUT

Meet the team
Contact us
Careers
FAQ

SERVICES

Data science & AI
Software engineering
Data engineering
Data strategy

OPEN SOURCE

[Cookiecutter Data Science](#)
[Deon: Ethics checklist](#)
[Zamba: Wildlife monitoring](#)
[Models from competitions](#)
[All open source projects](#)

COMPETITIONS

[Join a competition](#)
[Run a competition](#)
[How competitions work](#)
[Past competitions](#)

RESOURCES

[Blog](#)
[Projects](#)
[Insights](#)
[Case studies](#)

GOVERNMENT

[Contract Vehicles](#)
[Capability Statement](#)



Subscribe to our newsletter or follow us for updates

info@drivendata.org

[Terms](#)

[Privacy](#)

[Copyright policy](#)

© 2026 DrivenData LLC, 700 Colorado Blvd #611, Denver, CO 80206